Exercises

Exercise 1.1 · Algebraic form. Write in algebraic form:

1. $(3-2i)+(-1+5i)$;
2. $(3-2i)(-1+5i)$;
3. $(1-2i)^2$;
4. i^{2026}.

Solution.

1. $2+3i$.
2. $-3+15i+2i-10i^2=-3+17i+10=7+17i$.
3. $1-4i+4i^2=-3-4i$.
4. $2{,}026=4\times 506+2$, so $i^{2026}=i^2=-1$.

Exercise 1.2 · Inverses and quotients. Write in algebraic form: $\dfrac{1}{1+i}$; $\dfrac{2+i}{1-2i}$; $\dfrac{3+i}{2-i}$.

Solution. $\dfrac{1}{1+i}=\dfrac{1-i}{2}=\dfrac{1}{2}-\dfrac{1}{2}i$. Next $\dfrac{(2+i)(1+2i)}{5}=\dfrac{2+4i+i-2}{5}=i$. Finally $\dfrac{(3+i)(2+i)}{5}=\dfrac{6+3i+2i-1}{5}=1+i$.

Exercise 1.3 · Equations. Solve in $\mathbb{C}$:

1. $(1+i)z=3-i$;
2. $2z-3\overline{z}=1+10i$;
3. $z+2\overline{z}=6-i$.

Solution.

1. $z=\dfrac{3-i}{1+i}=\dfrac{(3-i)(1-i)}{2}=\dfrac{2-4i}{2}=1-2i$.
2. With $z=x+iy$: $2x+2iy-3x+3iy=-x+5iy$. Identification gives $-x=1$ and $5y=10$, that is $z=-1+2i$.
3. $x+iy+2x-2iy=3x-iy$, so $3x=6$ and $-y=-1$: $z=2+i$.

Exercise 1.4 · Real part and imaginary part of an expression. For $z=x+iy$ (x, y real), let $Z=z^2-2\overline{z}$.

1. Express $\mathrm{Re}(Z)$ and $\mathrm{Im}(Z)$ in terms of x and y.
2. Determine the complex numbers z for which Z is real.

Solution.

1. $z^2 = x^2 - y^2 + 2ixy$ and $-2\overline{z} = -2x + 2iy$. So $\mathrm{Re}(Z) = x^2 - y^2 - 2x$ and $\mathrm{Im}(Z) = 2xy + 2y = 2y(x+1)$.
2. Z is real if and only if $2y(x+1) = 0$, that is $y = 0$ or $x = -1$: z is real, or z has real part -1.

Exercise 1.5 · Binomial formula.

1. Using the binomial formula, write $(1+i)^5$ in algebraic form.
2. Deduce $(1+i)^{10}$.
3. Recover these two results from $(1+i)^2$.

Solution.

1. $(1+i)^5 = 1 + 5i + 10i^2 + 10i^3 + 5i^4 + i^5 = 1 + 5i - 10 - 10i + 5 + i = -4 - 4i$.
2. $(1+i)^{10} = (-4-4i)^2 = 16(1+i)^2 = 16 \times 2i = 32i$.
3. $(1+i)^2 = 2i$, so $(1+i)^4 = -4$, $(1+i)^5 = -4(1+i) = -4 - 4i$ and $(1+i)^{10} = (2i)^5 = 32i^5 = 32i$.

Exercise 1.6 · A purely imaginary quotient. Let z be a complex number such that $z\overline{z} = 1$ and $z \neq 1$. Let $w = \dfrac{1+z}{1-z}$.

1. Explain why $\overline{z} = \dfrac{1}{z}$.
2. Compute $\overline{w}$ and prove that w is purely imaginary.
3. Check the result for $z = i$.

Solution.

1. $z\overline{z} = 1$ forces $z \neq 0$, and $\overline{z} = \frac{1}{z}$.
2. By the properties of the conjugate, $\overline{w} = \dfrac{1+\overline{z}}{1-\overline{z}} = \dfrac{1+\frac{1}{z}}{1-\frac{1}{z}} = \dfrac{z+1}{z-1} = -w$. So w is purely imaginary.
3. $w = \dfrac{1+i}{1-i} = \dfrac{(1+i)^2}{2} = i$, which is indeed purely imaginary.

Exercise 1.7 · Sum of the powers of i.

1. Prove that, for every complex number $q \neq 1$ and every natural number n, $1 + q + \cdots + q^n = \dfrac{1 - q^{n+1}}{1 - q}$.
2. Deduce $S = 1 + i + i^2 + \cdots + i^{2026}$.
3. Recover this result by grouping the terms in blocks of four.

Solution.

1. If $S_n = 1 + q + \cdots + q^n$, then $S_n - qS_n = 1 - q^{n+1}$ (the other terms cancel). Since $1 - q \neq 0$, we divide. The proof only uses the rules of computation, which hold in $\mathbb{C}$.
2. $2{,}027 = 4 \times 506 + 3$, so $i^{2027} = i^3 = -i$ and $S = \dfrac{1 + i}{1 - i} = \dfrac{(1 + i)^2}{2} = i$.
3. Four consecutive powers add up to $i^{4k}(1 + i - 1 - i) = 0$. The 2,027 terms form 506 blocks with zero sum, and what remains is $i^{2024} + i^{2025} + i^{2026} = 1 + i - 1 = i$.

Problem

Problem · Square roots of a complex number. We look for the complex numbers z such that $z^2 = 3 + 4i$. Let $z = x + iy$, with x and y real.

1. Prove that $z^2 = 3 + 4i$ is equivalent to the system $x^2 - y^2 = 3$ and $2xy = 4$.
2. Prove that if $z^2 = 3 + 4i$, then $\left(x^2 + y^2\right)^2 = \left(x^2 - y^2\right)^2 + (2xy)^2$, and deduce that $x^2 + y^2 = 5$.
3. Deduce the two solutions of the equation, then check them.
4. By the same method, determine the complex numbers whose square is $-5 + 12i$.
5. Consider the equation $z^2 - 3z + 3 - i = 0$.
   (a) Prove that it can be written $\left(z - \frac{3}{2}\right)^2 = \frac{-3+4i}{4}$.
   (b) Determine the complex numbers w such that $w^2 = -3 + 4i$.
   (c) Deduce the solutions of the equation and check their sum and their product.

Solution · Problem.

1. $z^2 = x^2 - y^2 + 2ixy$. By uniqueness of the algebraic form, $z^2 = 3 + 4i$ is equivalent to $x^2 - y^2 = 3$ and $2xy = 4$.
2. The identity $\left(x^2 + y^2\right)^2 = \left(x^2 - y^2\right)^2 + 4x^2y^2$ is checked by expanding. It gives $\left(x^2 + y^2\right)^2 = 9 + 16 = 25$, and since $x^2 + y^2 \geq 0$, $x^2 + y^2 = 5$.
3. Adding and then subtracting $x^2 - y^2 = 3$: $x^2 = 4$ and $y^2 = 1$. Since $xy = 2 > 0$, x and y have the same sign: $z = 2 + i$ or $z = -2 - i$. Check: $(2 + i)^2 = 4 + 4i - 1 = 3 + 4i$, and $(-2 - i)^2 = (2 + i)^2$.
4. $x^2 - y^2 = -5$, $xy = 6$ and $x^2 + y^2 = \sqrt{25 + 144} = 13$. So $x^2 = 4$, $y^2 = 9$, with x and y of the same sign: $z = 2 + 3i$ or $z = -2 - 3i$. Check: $(2 + 3i)^2 = 4 + 12i - 9 = -5 + 12i$.
5. (a) $\left(z - \frac{3}{2}\right)^2 = z^2 - 3z + \frac{9}{4}$, so the equation can be written $\left(z - \frac{3}{2}\right)^2 = \frac{9}{4} - 3 + i = \frac{-3+4i}{4}$.
   (b) $x^2 - y^2 = -3$, $xy = 2$, $x^2 + y^2 = 5$: $x^2 = 1$, $y^2 = 4$, hence $w = 1 + 2i$ or $w = -1 - 2i$.
   (c) $z - \frac{3}{2} = \pm\frac{1+2i}{2}$, hence $z_1 = 2 + i$ and $z_2 = 1 - i$. Their sum is 3 and their product is $(2 + i)(1 - i) = 2 - 2i + i + 1 = 3 - i$, as the coefficients of the equation predict.

Teacher note · Status of the problem. Complex square roots and quadratic equations with complex coefficients are "possible problems" of the syllabus, not required. Question 2 avoids the modulus, introduced in the next chapter: one can come back to it by noting that $x^2 + y^2 = |z|^2 = |z^2|$.